Randall D. Kamien Fall 2026

Physics 6611: Statistical Mechanics

This course is intended to be an introductory graduate course on statistical mechanics.

Topics to be covered

1. Hamiltonian Dynamics, Phase Space Flows, Liouville's Theorem
2. The Ergodic Theorem and the Microcanonical Ensemble
3. The Canonical Ensemble
4. The Ideal Gas, Indistinguishability and Planck's Constant
5. Legendre and Laplace Transforms, Thermodynamics
6. The Grand Canonical Ensemble; The Gibbs Phase Rule
7. The Cluster Expansion; The Virial Expansion
8. Quantum Statistical Mechanics; Fermions and Bosons
9. The Ising Model
10. High Temperature Expansions
11. Low Temperature Expansions
12. Introduction to Phase Transitions and Critical Phenomena
13. Other Spin Models
14. Universality
15. Random Walks
16. Path Integrals
17. The Diffusion Equation
18. The Fokker-Planck and Smoluchowski Equations
19. The Black-Scholes Option Pricing Model

Prerequisites: Quantum Mechanics at the Level of Baym
Course Times: MW: 10:15AM-11:45AM **Location**: DRL 3C6
Text: Pathria, *Statistical Mechanics*
Homework & Exams: There will be 4 homework assignments which will be spaced uniformly in time along the course and will be due one week after they are assigned. Since we will post solutions online, no homework will be accepted late. Grade entirely based upon homework. No exams.
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